

Transforming Pitching: A Data-Driven Approach to Mechanics and Coaching Using Deep Learning

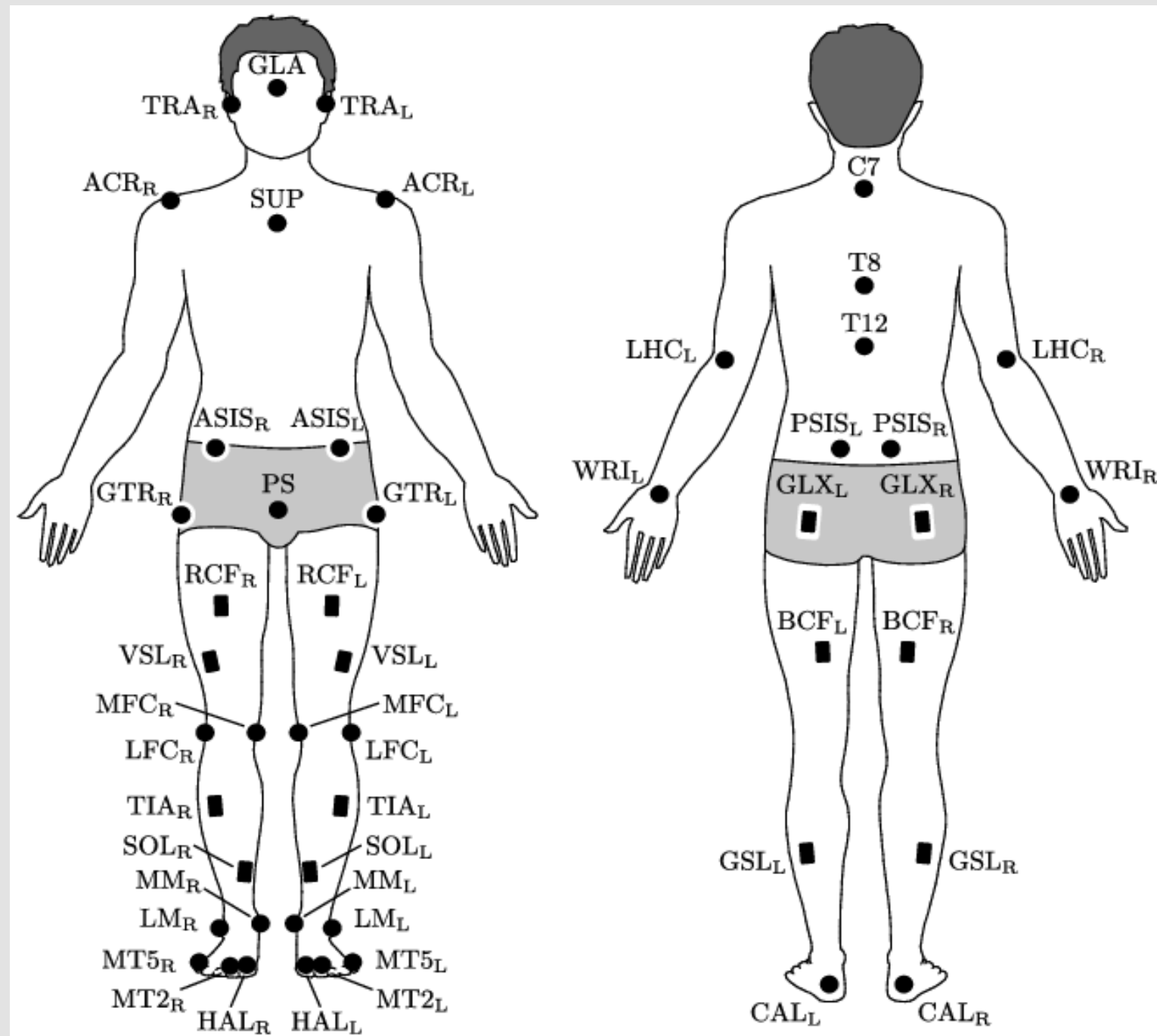
Caleb Guisewite

Belmont University, Nashville Tennessee

Introduction

Pitching is one of the most biomechanically demanding movements in sports, where small variations in mechanics can have major impacts on velocity, performance, and injury risk. This project explores the ability of deep learning models — particularly transformer-based architectures — to learn the complex, time-dependent relationships that underlie pitch velocity. By comparing machine learning models using static data with advanced deep learning models trained on sequential motion capture data, we aim to uncover not only how fast a pitcher throws, but why — providing interpretable insights that could inform training, injury prevention, and player development.

Data and Feature Overview



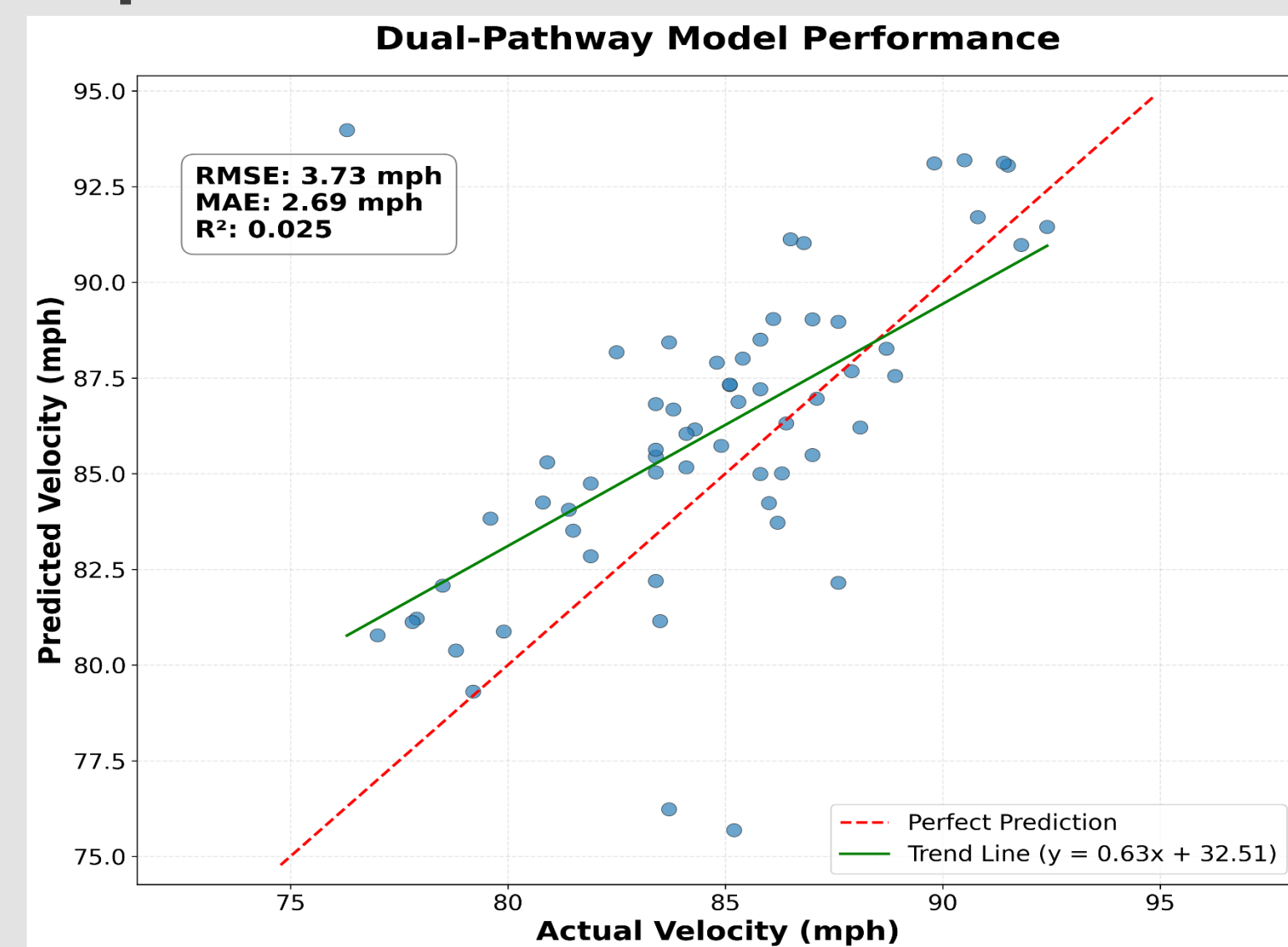
The full sequential biomechanical data is divided up into 6 groups

- **Energy Flow:** energy transfer, generation, and absorption
- **Force Plate:** Rear leg, lead leg ground reaction forces
- **Force Moments:** Joint Forces and Moments
- **Joint Angles:** Joint angles
- **Joint Velos:** Joint angular velocities
- **Landmarks:** Joint Center Positions

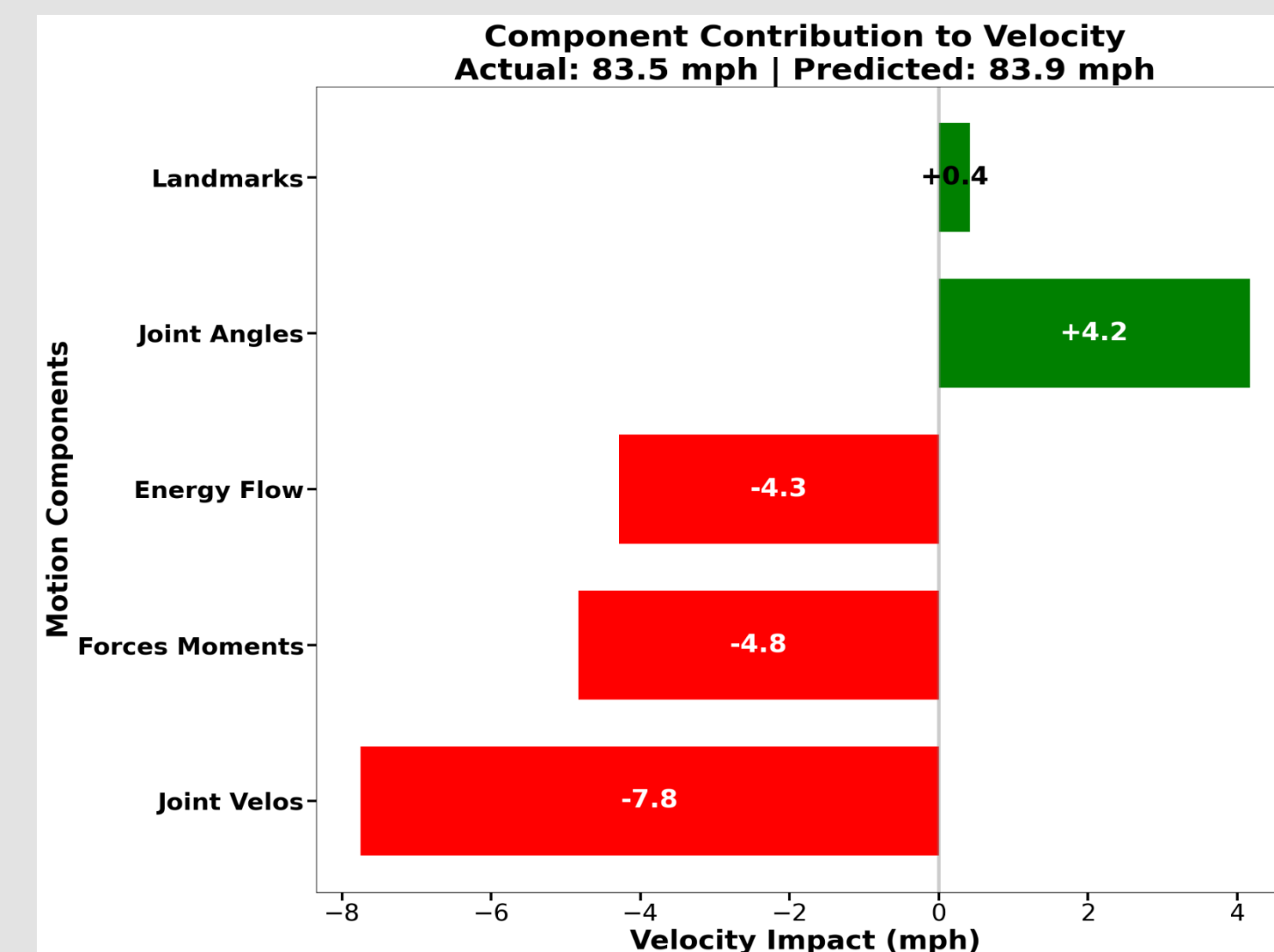
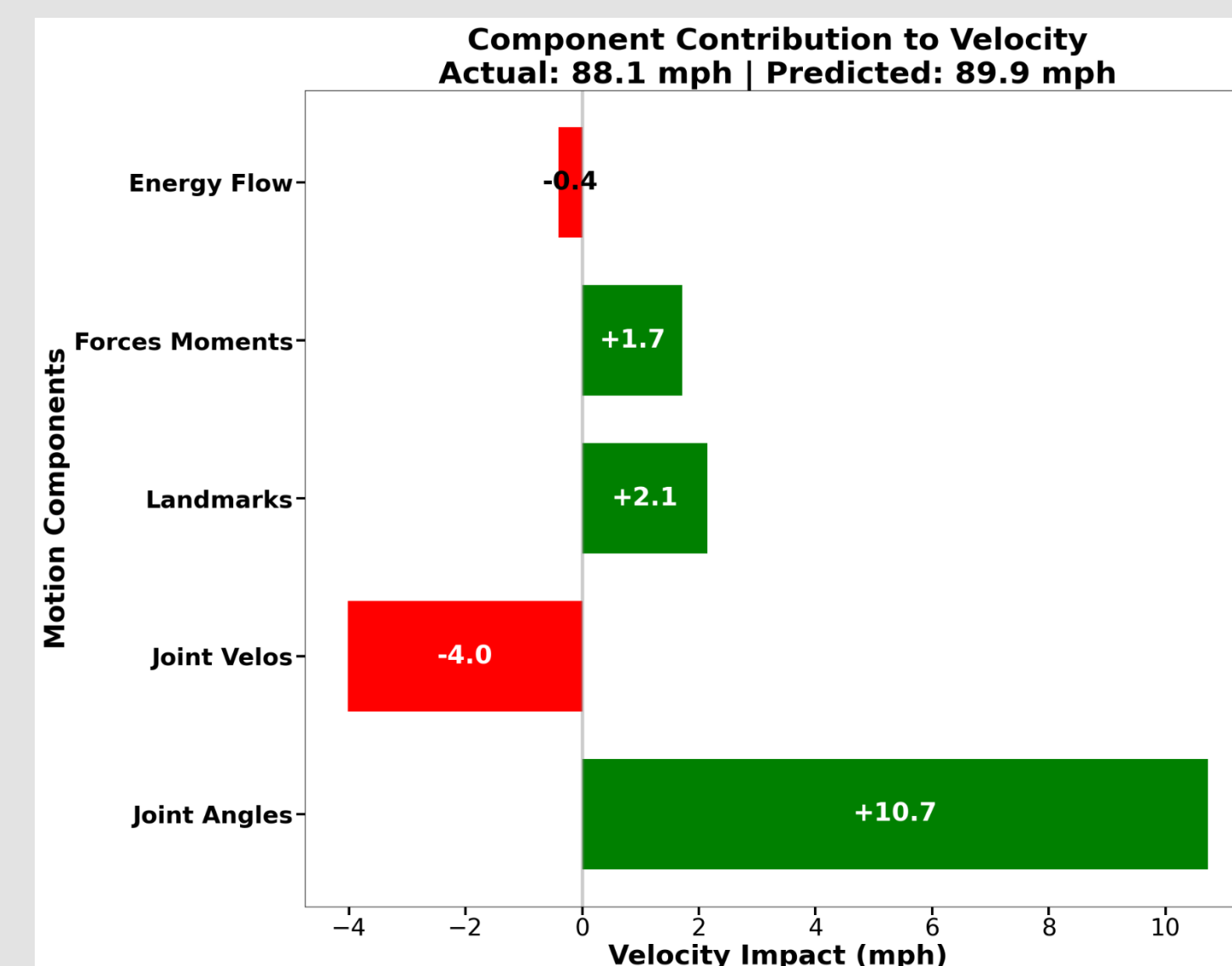
Next, we have static data

- **Point of Interest Data:** Kinematic, kinetic, and energetic metrics commonly referenced in biomechanical analyses

Sequential Data Performance

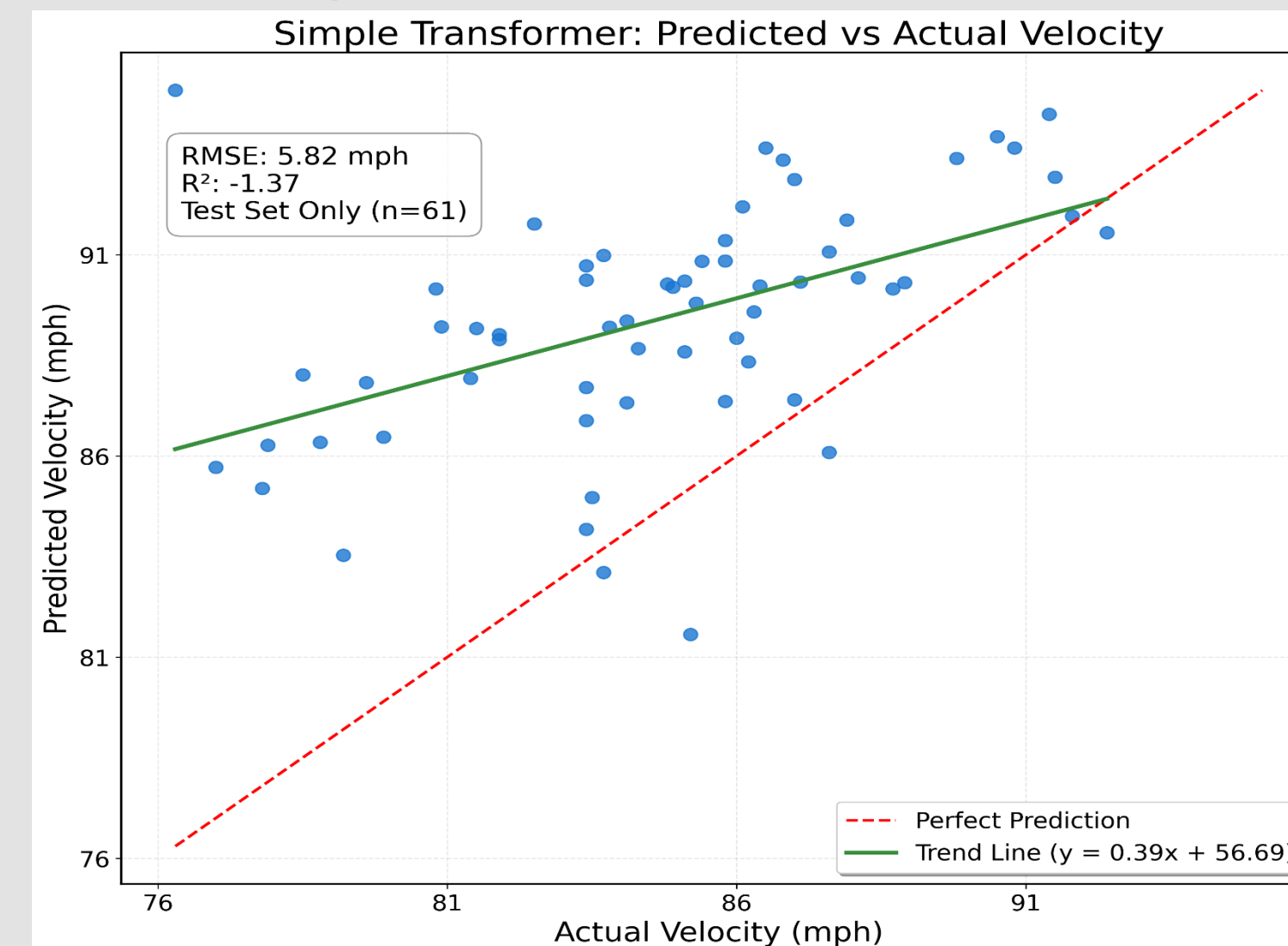


Individual Athlete Insights

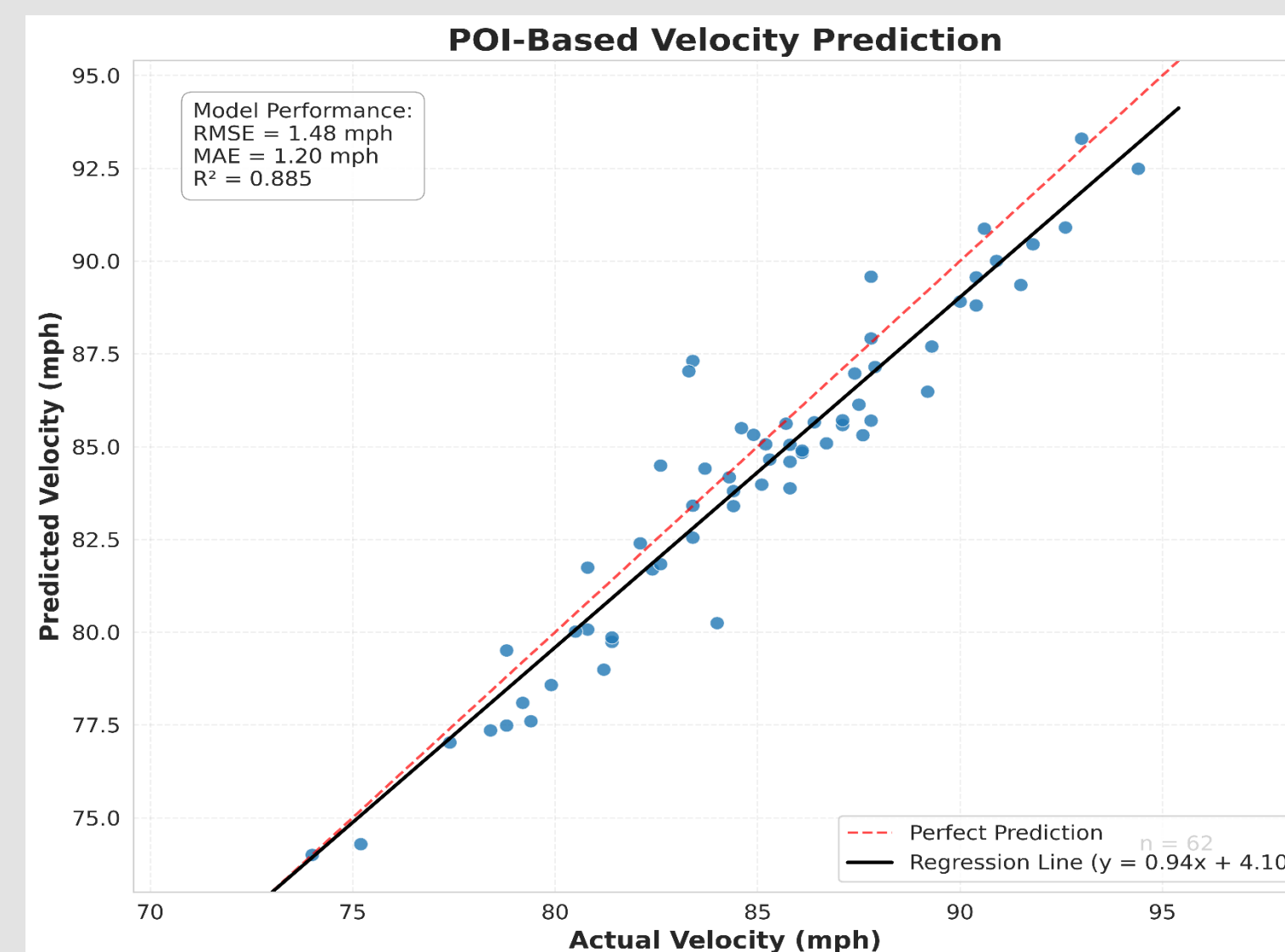


Individual athlete's biomechanical component contribution toward velocity.

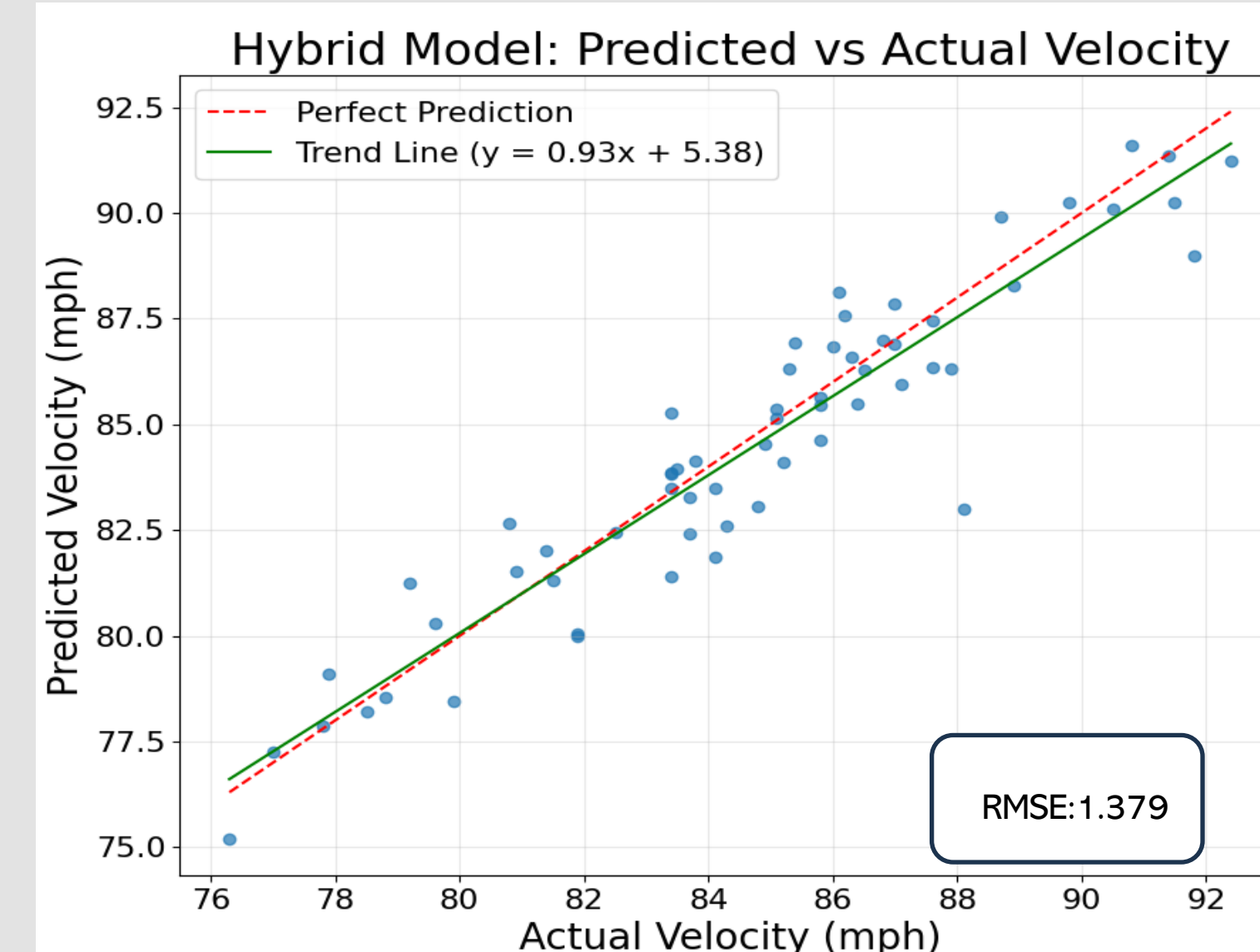
Comparing Models



Simple transformer model trained on sequential biomechanical data.

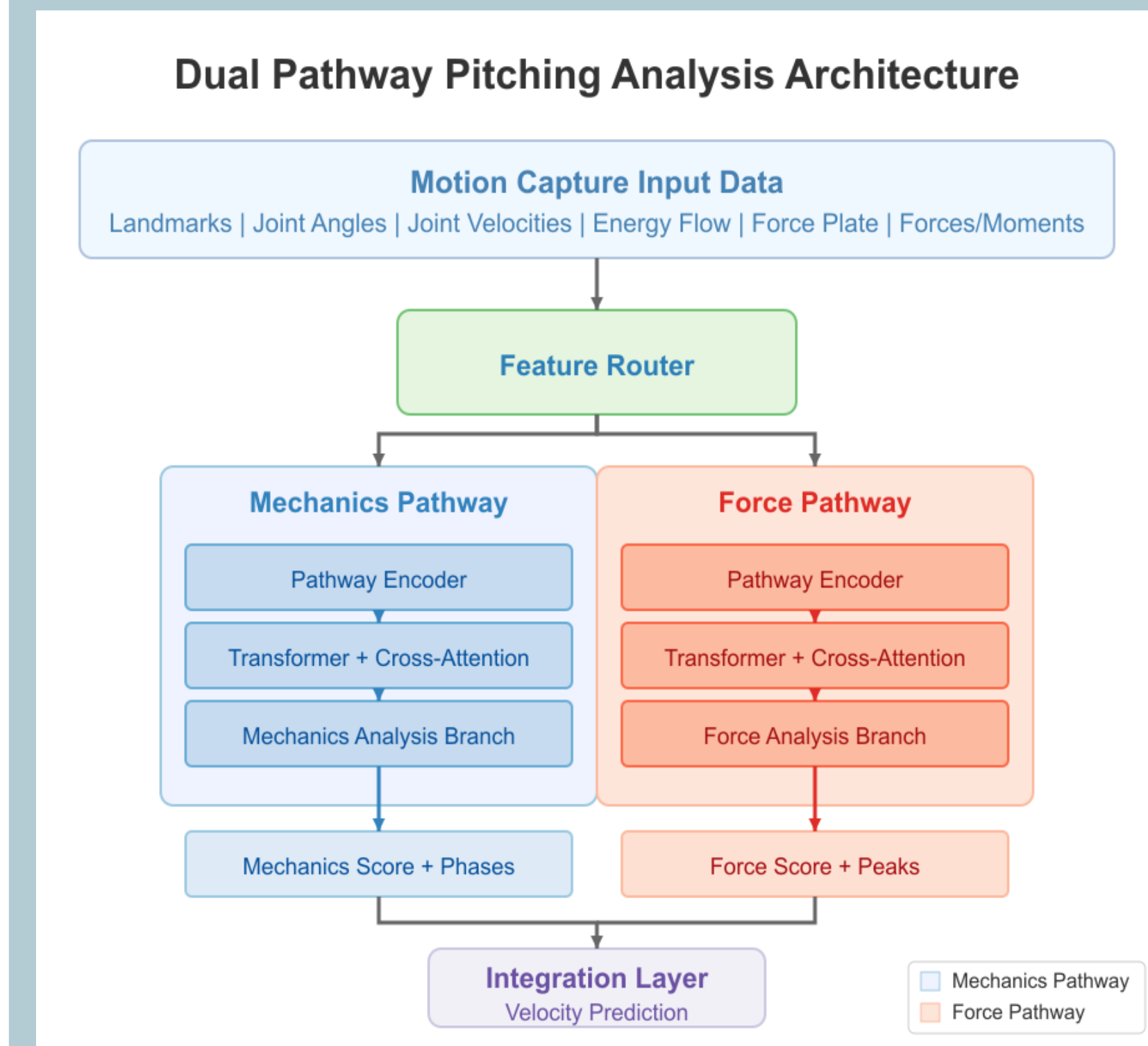


Neural network-based model trained on static POI Data



Hybrid transformer model trained on POI & sequential biomechanical data.

Model Architecture



Future of Baseball Analytics

As data collection in baseball evolves from stat sheets to high-resolution biomechanics, the next frontier in analytics lies not just in what happened, but how and why. Motion capture systems, force plates, and wearable tech now provide detailed measurements of an athlete's movement, enabling teams to analyze the mechanics behind every pitch, swing, and sprint.

Incorporating biomechanics into predictive models opens the door to deeper, more personalized insights. Rather than relying solely on historical outcomes, teams can now project performance based on how a player moves — offering early signals of potential, inefficiency, or injury risk. Deep learning models, especially transformers, are uniquely positioned to harness these complex time-series inputs. They can learn hidden dependencies across joints, timing, and force application that traditional models overlook.

I believe that the future of baseball analytics lies within biomechanical analysis through prediction and simulation of execution by players.